Requirements Gathering Analysis: Disaster Alert & Resource Allocation System

Introduction

Early warning and emergency relief are vital public services that directly influence safety, survival and the speed of recovery after a disaster. Despite advances in technology, many civil defense operations still rely on manual workflows such as phone calls, paper lists and a few broadcast channels. The Disaster Alert & Resource Allocation System initiative aims to modernize these operations and address late warnings, duplicated or misdirected relief, and the lack of a single live picture of the situation.

This study gathers, synthesizes and formalizes the requirements for a web-based platform. Its primary objectives are:

- **To analyze the operational bottlenecks in the current manual alert and resource allocation workflows.**
- **To investigate user perceptions, emotional stress points and unmet needs among key stakeholders.**
- **To define the core functional, operational and architectural requirements of the proposed system.**

Methodology

A qualitative research design was used, consisting of in-depth semi-structured interviews with three stakeholder groups: Emergency Operators, Relief Agencies and Citizens. The findings were categorized across four dimensions: Current State of the System, Interviewee Opinions, Interviewee Feelings, and System Goals.

Findings

1. Current State Analysis

The baseline investigation showed that alerting and relief coordination depend on phone calls, paper lists and a few broadcast channels.

- **Alert Distribution Bottlenecks: Warnings about floods, fires, earthquakes and storms reach people late, or only through TV and word of mouth, with no targeting by area or severity.**
- **Resource Assignment Workflow: Emergency teams, shelters, medical supplies, food and vehicles are assigned by phone and paper lists. This causes misdirected or duplicated deliveries and manual counting errors.**
- **Lack of Visibility: There is no single live picture of the hardest-hit areas, what each shelter or depot holds, or which needs are still unmet. Records can be lost when time matters most.**

2. Stakeholder Perceptions

- **Alert Speed and Reach: Citizens report late or overly general warnings. Authorities acknowledge that current channels cannot guarantee everyone is reached, and support multi-channel delivery (SMS, email, push) targeted by area and severity.**
- **Resource Allocation Efficiency: Relief agencies see supplies wasted through duplication while other areas go short. Management agrees that a live inventory dashboard with allocation suggestions based on severity and population would improve fairness and accuracy.**

3. User Experience & Psychosocial Dynamics

- **Crisis Atmosphere: Citizens feel fear and anxiety because they lack reliable information about the danger and the nearest shelter. Staff feel overwhelmed distributing limited resources without a clear picture, and fear errors when every minute matters.**
- **Timeliness of Warnings: Interviewees stated that current warnings do not reach citizens in affected areas in time to act.**

System Requirements

To address the identified shortcomings, the proposed platform must provide the following capabilities:

- **Multi-Channel Alerting: Authorities publish alerts by area and severity and deliver them to citizens by SMS, email and push notification within minutes.**

- **Live Resource Inventory: Real-time tracking of ambulances, rescue teams, water, food, medicine and shelter beds, which prevents duplicated or missed deliveries.**
- **Allocation Suggestions and Dashboard: An interactive map and dashboard showing affected zones and unmet needs, with allocation suggestions based on the severity and population of each zone, plus reports and statistics for decision makers.**
- **Security, Access Control and Availability: Role-based access so only authorized staff can issue alerts, protection of citizens' personal data (phone numbers, locations), integration with SMS gateways and weather or seismic data feeds, and high availability under heavy load and weak mobile networks.**

Conclusion

The qualitative findings show that replacing phone calls and paper lists with an integrated digital platform is necessary for fast warning, efficient use of limited resources and better decisions. Turning manual workflows into real-time digital processes addresses the core bottlenecks, reduces response time and prevents duplicated or missed deliveries.

Recommendations

1. **Phased System Deployment: Implement multi-channel alert publishing and the live resource inventory first, followed by the allocation suggestion engine and the full analytics dashboard.**
2. **Stakeholder Onboarding: Run targeted training and drills for emergency operators and relief staff on alert issuing, the dashboard and reporting tools.**
3. **Continuous Improvement: Use post-incident reports and statistics to evaluate response times, adjust allocation rules and strengthen preparedness before each disaster season.**